ALGORITHMS II

Week 2

REVISION

ACTIVITY

BATH FLOWCHART

Display the following as a flowchart. You may need to change the order or add actions.

1. Put in the plug.
2. Fill the bath to the correct level.
3. Check the temperature is OK.

PSEUDOCODE

a structured, code-like language that can be used to describe an algorithm.

MORE ABOUT PSEUDOCODE

- It allows the **developer** to concentrate on the **logic** and efficiency of the algorithm without having to bother about the rules of any programming language. It is relatively straightforward to translate an algorithm written in pseudocode into any high-level programming language.

Developer: a person whose job it is to create new software

Logic: the principles and reasoning underlying the constructs and elements to be applied in solving problems

ALGORITHM FOR ADDING TWO NUMBERS

Written description:

1. Enter first number.
2. Enter second number.
3. Calculate total by adding first and second numbers.
4. Output total.

ALGORITHM FOR ADDING TWO NUMBERS

Flowchart:



▲ Figure 1.4 Flowchart showing the adding of two numbers

ALGORITHM FOR ADDING TWO NUMBERS

PSEUDOCODE:

```
SEND 'Please enter the first number' TO DISPLAY
RECEIVE firstNumber FROM KEYBOARD
SEND 'Please enter the second number' TO DISPLAY
RECEIVE secondNumber FROM KEYBOARD
SET total TO firstNumber + secondNumber
SEND total TO DISPLAY
```

SUBJECT VOCABULARY

VARIABLE

a 'container' used to store data. The data stored in a variable is referred to as a value.

IMPORTANT: The value stored in a variable is **not fixed**. The same variable can store different values during a program and each time a program is run

IDENTIFIER

- A **unique** name given to a variable or a constant.

Using descriptive

names for variables makes code much

easier to read.

ARITHMETIC OPERATOR

- an operator that performs a calculation on two numbers

CONSTANT

- a 'container' that holds a value that never changes.

like variables, constants have unique identifiers

ARITHMETIC OPERATORS

OPERATOR	FUNCTION	EXAMPLE
+	Addition: add the values together.	8 + 5 = 13 myScore1 + myScore2
–	Subtraction: subtract the second value from the first.	17 – 4 = 13 myScore1 – myScore2
*	Multiplication: multiply the values together.	6 * 9 = 54 numberBought * price
/	Real division: divide the first value by the second value and return the result including decimal places.	13 / 4 = 3.25 totalMarks/numberTests
DIV	**Quotient**: like division, but it only returns the whole number or *integer*.	13 DIV 4 = 3 totalMarks DIV numberTests
MOD	**Modulus**/modulo: this will return the remainder of a division.	13 / 4 = 3 remainder 1 Therefore 13 MOD4 = 1
^	Exponentiation: this is for 'to the **power** of'.	3 ^ 3 = 27 It is the same as writing 3^3

CHECK PAGES (7 – 9)